IT079IU — Lab 2: Introduction to Relational Algebra

International University – VNU-HCM · School of Computer Science and Engineering · IT079IU Principles of Database Management

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Lab number: Lab 2

1. Dataset and conventions

Tool: RelaX relational algebra calculator. Database: **Employee (Franklin University)**, selected via "Select DB" in RelaX. Each query below was executed in RA mode with "Execute query".

Relation	Attributes
EMPLOYEE	Fname, Minit, Lname, Ssn, Bdate, Address, Sex, Salary, Super_ssn, Dno
DEPARTMENT	Dname, Dnumber, Mgr_ssn, Mgr_start_date
PROJECT	Pname, Pnumber, Plocation, Dnum
WORKS_ON	Essn, Pno, Hours
DEPENDENT	Essn, Dependent_name, Sex, Bdate, Relationship

Department numbers appear as **Dno / Dnumber / Dnum**, person identifiers as **Ssn / Essn / Mgr_ssn / Super_ssn**, and project numbers as **Pnumber / Pno**. The names therefore never coincide across relations, so a natural join would miss: every join below is a theta join with an explicit condition. "Names" means **Fname, Lname** separately. Renaming is written old → new (**->**) throughout. String literals use single quotes as in the RelaX help (`sigma firstname = 'Bob' (Customer)`). Numeric attributes (Ssn, Salary, Dno, …) take unquoted literals — quoting a number raises a type error. Query 5 uses the grouping operator γ, which is extended relational algebra beyond the five basic operators.

2. Practice exercises

Q1 — Names of all employees.

π Fname,Lname(EMPLOYEE)

```
pi Fname, Lname (EMPLOYEE)
```

Q2 — Names of all female employees.

π Fname,Lname(σ Sex='F'(EMPLOYEE))

```
pi Fname, Lname (sigma Sex='F' (EMPLOYEE))
```

Q3 — Employee names with department names.

π Fname,Lname,Dname(EMPLOYEE ⋈ EMPLOYEE.Dno=DEPARTMENT.Dnumber DEPARTMENT)

```
pi Fname, Lname, Dname (EMPLOYEE join EMPLOYEE.Dno = DEPARTMENT.Dnumber
(DEPARTMENT))
```

Q4 — Department names with manager's SSN.

π Dname,Mgr_ssn(DEPARTMENT)

```
pi Dname, Mgr_ssn (DEPARTMENT)
```

Q5 — Average salary in each department (extended RA).

γ Dno; AVG(Salary)→AvgSalary(EMPLOYEE), and grouped by Dname after the join for department names.

```
gamma Dno; AVG(Salary)->AvgSalary (EMPLOYEE)
gamma Dname; AVG(Salary)->AvgSalary (EMPLOYEE join EMPLOYEE.Dno =
DEPARTMENT.Dnumber (DEPARTMENT))
```

Q6 — Employees who work on no projects.

π Fname,Lname(EMPLOYEE ⋈ EMPLOYEE.Ssn=Nowork.Ssn ρNowork(π Ssn(EMPLOYEE) − ρ Ssn←Essn(π Essn(WORKS_ON))))

```
pi Fname, Lname (EMPLOYEE join EMPLOYEE.Ssn = Nowork.Ssn (rho Nowork (pi Ssn
(EMPLOYEE) - (pi Essn -> Ssn (WORKS_ON)))))
```

Live result: empty — every employee in this dataset works on at least one project.

Q7 — Employees and their dependents.

Inner join (only employees with dependents): π Fname,Lname,Dependent_name(EMPLOYEE ⋈ EMPLOYEE.Ssn=DEPENDENT.Essn DEPENDENT). For all employees including those without dependents, use a left outer join instead.

```
pi Fname, Lname, Dependent_name (EMPLOYEE join EMPLOYEE.Ssn = DEPENDENT.Essn
(DEPENDENT))
```

Q8 — Employees who are also department managers.

π Fname,Lname(EMPLOYEE ⋈ EMPLOYEE.Ssn=DEPARTMENT.Mgr_ssn DEPARTMENT)

```
pi Fname, Lname (EMPLOYEE join EMPLOYEE.Ssn = DEPARTMENT.Mgr_ssn (DEPARTMENT))
```

Q9 — Employees who work on every project (division).

π Fname,Lname(EMPLOYEE ⋈ EMPLOYEE.Ssn=WorkAll.Essn ρWorkAll(π Essn,Pno(WORKS_ON) ÷ ρPnumber→Pno(π Pnumber(PROJECT))))

```
pi Fname, Lname (EMPLOYEE join EMPLOYEE.Ssn = WorkAll.Essn (rho WorkAll ((pi
Essn, Pno (WORKS_ON)) / (rho Pnumber -> Pno (pi Pnumber (PROJECT))))))
```

Live result: empty — no one works on all 6 projects (the most is 4, by Franklin Wong, Ssn 333445555).

Q10 — Same supervisor as employee 123456789 (excluding that employee; drop the Ssn != clause to include them).

π Fname,Lname(σ Super_ssn=Sup ∧ Ssn≠123456789(EMPLOYEE × ρS(π Super_ssn→Sup(σ Ssn = 123456789(EMPLOYEE)))))

```
S = pi Super_ssn -> Sup (sigma Ssn = 123456789 (EMPLOYEE))
pi Fname, Lname (sigma Super_ssn = Sup AND Ssn != 123456789 (EMPLOYEE cross join S))
```

Drop the `Ssn != ` clause to include 123456789 itself. Live result: Ramesh Narayan and Joyce English (supervisor 333445555).

3. Execution screenshots

Queries were executed in RelaX (Employee database); the operator tree and result table were checked for each. Representative screenshots:

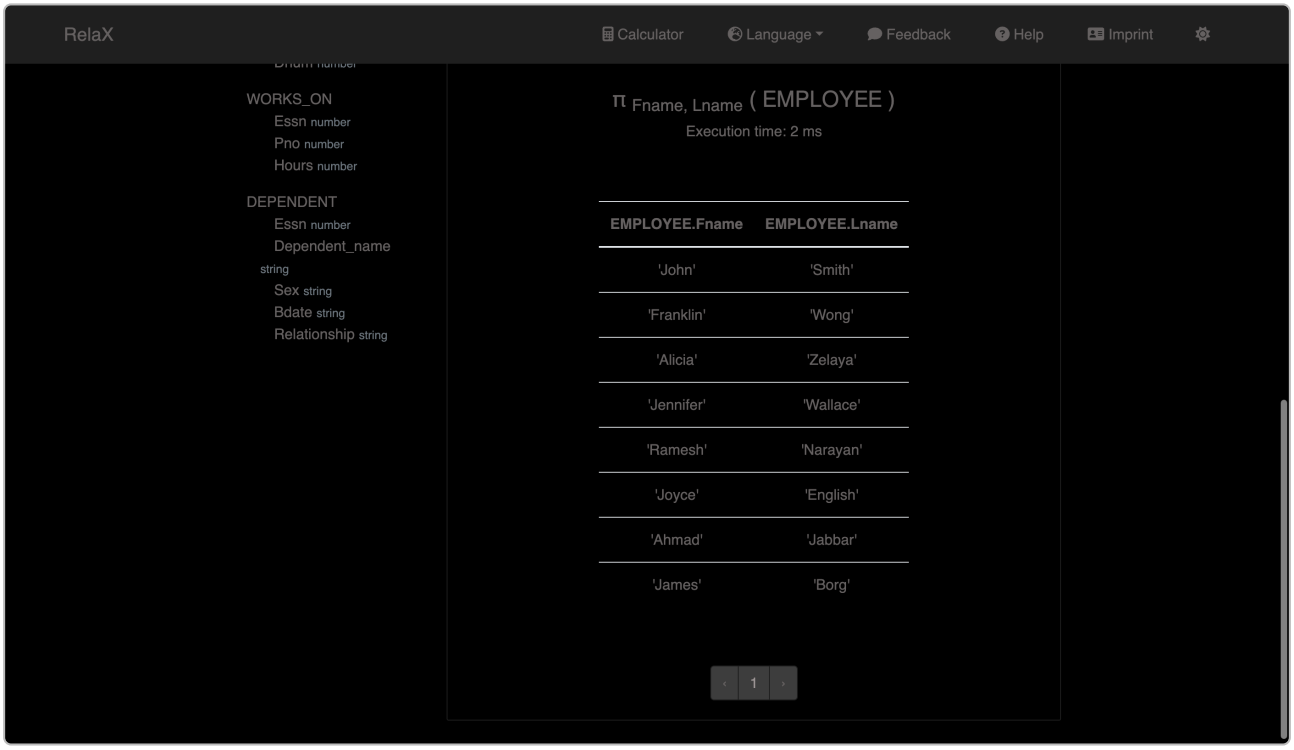


Fig. 1 — Q1: projection of all employee names, 8 rows.

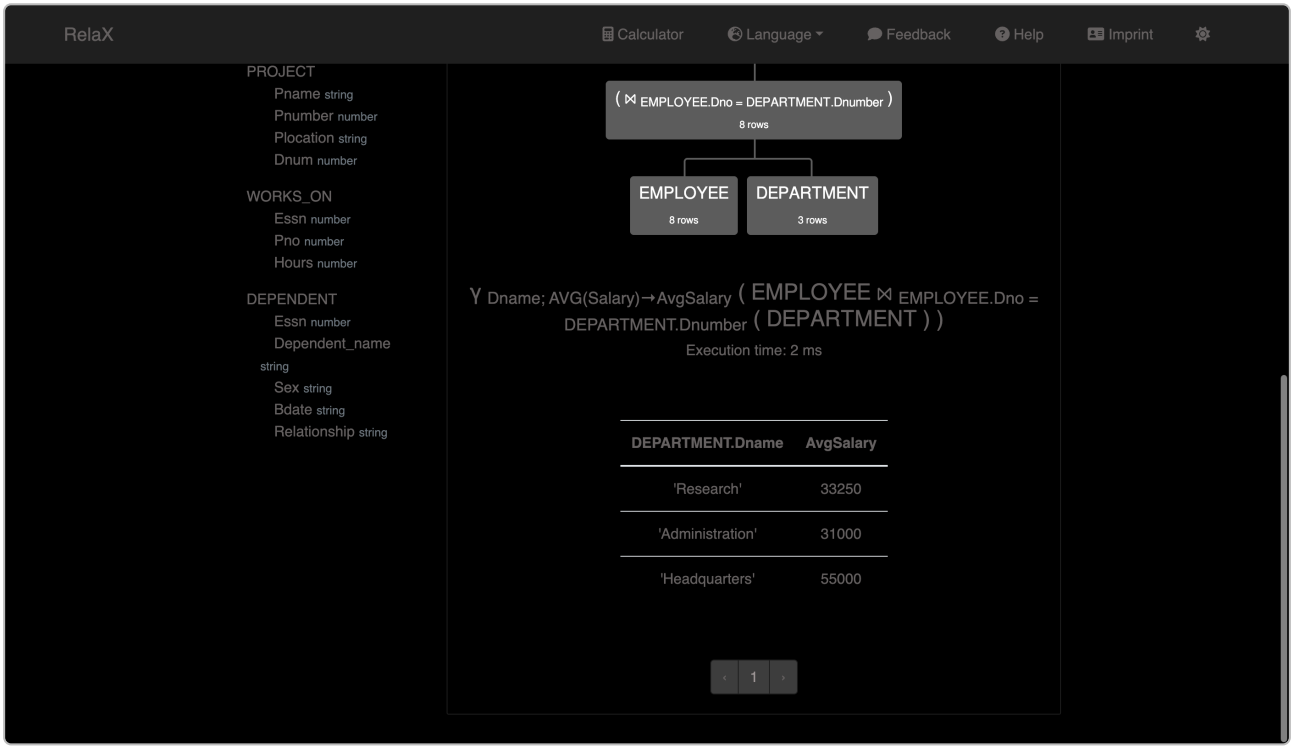


Fig. 2 — Q5: average salary grouped by department name (extended RA γ).

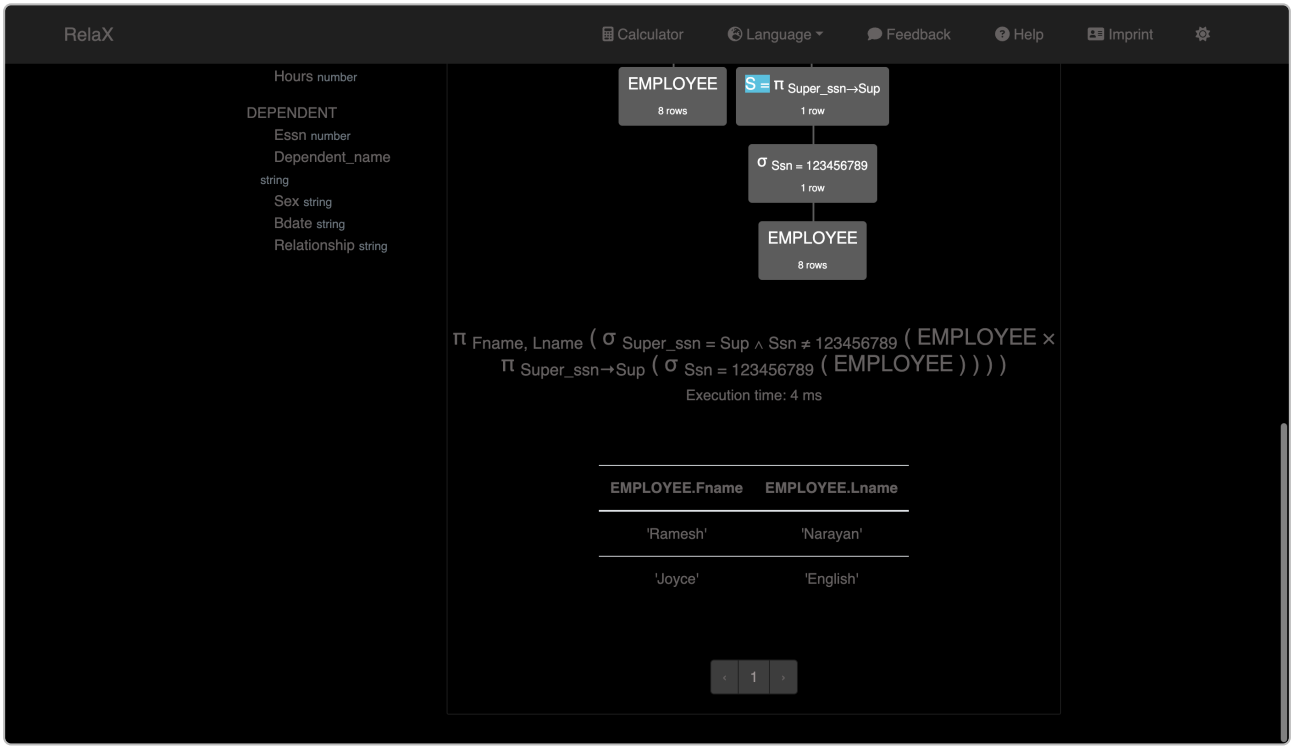


Fig. 3 — Q10: employees sharing the supervisor of 123456789 (Ramesh Narayan, Joyce English).

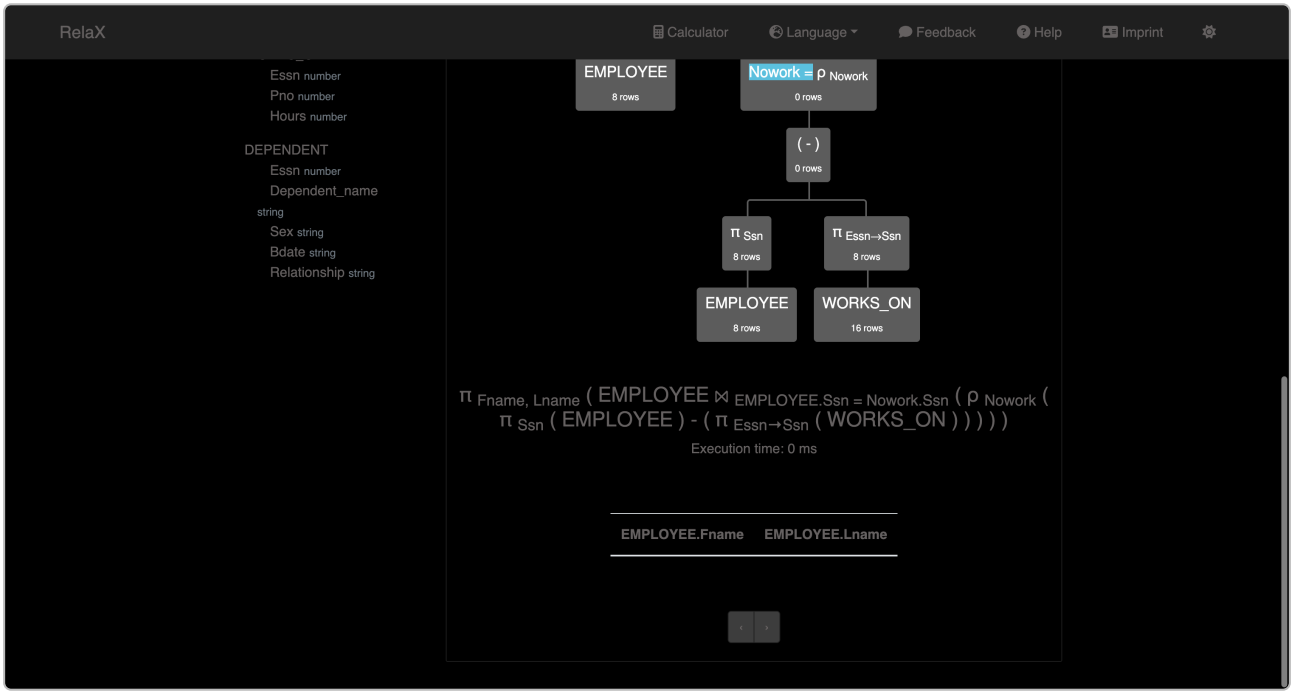


Fig. 4 — Q6: difference yields an empty Nowork set (0 rows), so the result table is empty — every employee works on at least one project.

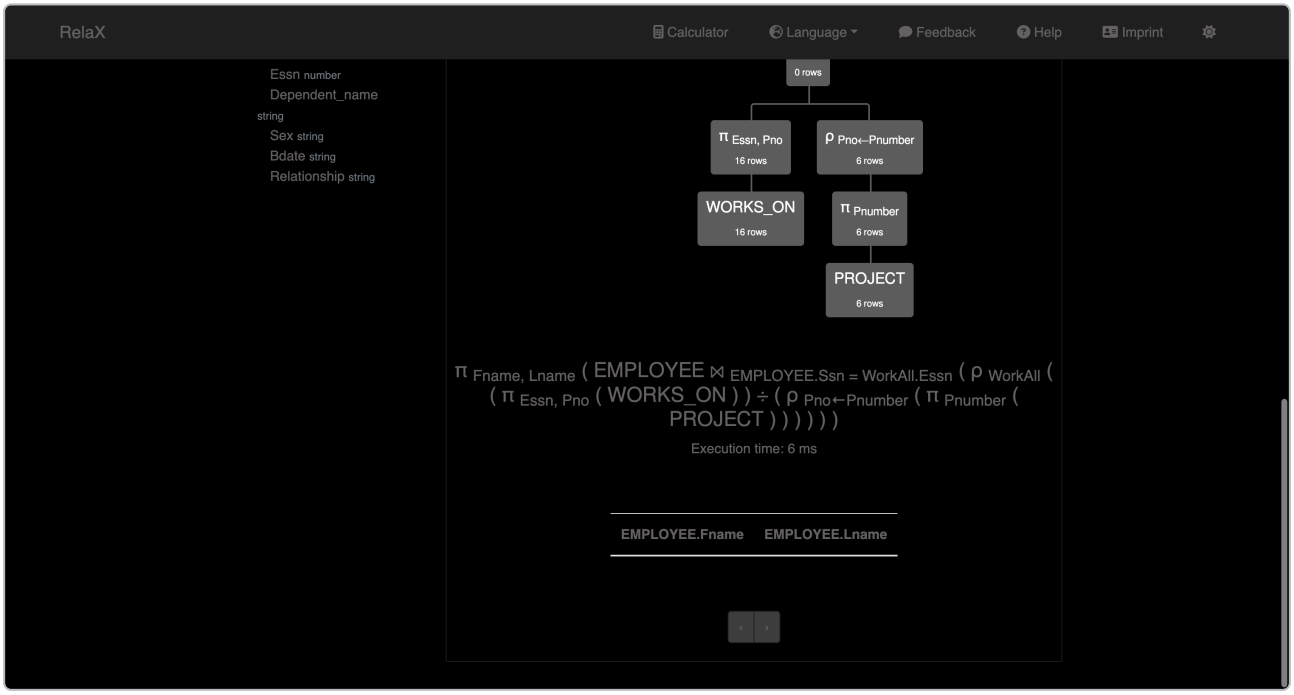


Fig. 5 — Q9: division yields an empty WorkAll set (0 rows), so the result table is empty — nobody works on all 6 projects.

Appendix — Lab2_Data.txt worked examples (corrected)

The guide omits the join conditions, and example (c) as written subtracts relations of different arity, which is not union-compatible. Corrected versions (verified by pasting the data file into the Group Editor; live results: a → all 3 courses; b → B, C, D; c → A only):

```
a. pi CourseName (Course join Course.CourseId = Register.CourseID (Register))
b. pi StudentName (sigma CourseID='IT162IU' (Student join Student.StudentID =
Register.StudentID (Register)))
c. pi StudentName (Student join Student.StudentID = Nowork.StudentID (rho Nowork
((pi StudentID (Student)) - (pi StudentID (Register)))))
```

AI use: An AI assistant drafted the relational algebra expressions and this report text. The queries were executed and verified by me in RelaX, the design decisions were approved by me, and I am able to justify every query.